Lucy Gow

Personal Details

Nationality:	Australian
Postal Address:	Max-Planck-Institut für Gravitationsphysik Am Mühlenberg 1, 14476 Potsdam-Golm Germany
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Research Experience

Junior Scientist October 2008 -
Max Planck Institute for Gravitational Physics, Potsdam
Topic: Aspects of Yangians and quantum superalgebras in connection with integrable models and string/gauge dualities.

Research Associate February 2008 - September 2008
University of Sydney, Supervisor: Alexander Molev
Topic: Representations of the twisted quantum Yangian of type C

Education

PhD (Mathematics)
University of Sydney, Australia, 2008
Thesis title: Yangians of Lie Superalgebras.
Supervisor: Alexander Molev; Associate Supervisor: Ruibin Zhang.

Bachelor of Science (Mathematics) - Honours
First Class Honours, University of Sydney, Australia, 2004

Bachelor of Science/Bachelor of Arts (Linguistics)
University of New South Wales, Australia, 2002

Publications

- Gauss Decomposition of the Yangian $Y(\mathfrak{gl}_{m|n})$, *Communications in Mathematical Physics* 276(3) December 2007, pp 799-825.
- On the Yangian $Y(\mathfrak{gl}_{m|n})$ and Its Quantum Berezinian, *Czechoslovak Journal of Physics* 55(11) November 2005, pp 1415-1420.

Conference Presentations and Seminars

- *Gauss decomposition of the super-Yangian* $Y(\mathfrak{gl}_{m|n})$, University of Sydney Algebra Seminar, 2008
- *Introduction to Hopf Algebras*, Monash University Colloquium, Melbourne 2008,
- *What is a Yangian?*, Melbourne University Mathematics Seminar, Melbourne 2008,
- *Yangians of Lie Superalgebras*, Australian Mathematical Society Meeting, Macquarie University, Sydney 2006
- *The Yangian* $Y(\mathfrak{gl}_{m|n})$ *and its Centre*, Australian Mathematical Society Meeting, UWA, Perth 2005
- *The Yangian* $Y(\mathfrak{gl}_{m|n})$ *and its Centre*, International Colloquium on Integrable Systems and Quantum Symmetries, Prague 2005
- *Yangians, Quantum Determinants and Quasideterminants*, Victorian Algebra Conference, RMIT, Melbourne 2004.

Conference and Workshop Participation

- Algebraic Lie Structures with Origins in Physics, Isaac Newton Institute for Mathematical Sciences, Cambridge UK, 23-27 March 2009
- Workshop on Representations and Cohomology, University of Cologne, 1-6 March, 2009
- Geometry and Integrability, University of Melbourne, 6-15 February 2008
- Geometry and Lie Theory, University of Sydney, July 2007
- Annual Meeting of the Australian Mathematics Society, Macquarie University, 2006
- ICE-EM Graduate Winter School, University of Queensland, 2006.
- QFT and Its Ramifications, University of Tasmania, 2005
- Annual Meeting of the Australian Mathematics Society, University of Western Australia, 2005
- Victorian Algebra Conference, University of Western Australia, 2005
- Integrable Systems and Quantum Symmetries, Prague, 2005
- Lie Algebras, Vertex Operator Algebras and Their Applications, North Carolina State University, 2005
- Mathematical Physics and Lie Theory Workshop, University of Queensland, November-December 2004
- Annual Meeting of the Australian Mathematics Society, RMIT, 2004
- Victorian Algebra Conference, RMIT, 2004.

Prizes and Scholarships

- Australian Postgraduate Award, University of Sydney, 2004-2007
- School of Mathematics Supplementary Award, University of Sydney 2004-2007
- Australian Federation of University Women NSW Prize, University of Sydney, 2003
- Faculty of Science Honours Scholarship, University of Sydney, 2003
- School of Mathematics Vacation Scholarship, University of NSW, 2001
- Head of Schools Prize for excellence in Level 2 Mathematics, University of NSW, 1999
- UMAP Exchange Scholarship, University of NSW/Yonsei University, Korea, 1999

Teaching Experience

Lecturer - MATH 1001 Differential Calculus
University of Sydney Summer School, 2007
- Coordinated tutorials
- Wrote and graded assessment materials including the final exam
- Created and maintained a course website

Tutor
School of Mathematics and Statistics, University of Sydney, 2004-2006
- Subjects taught:
  - ◇ Linear Algebra (three times)
  - ◇ Discrete Mathematics
  - ◇ Calculus for Life Sciences (twice)
  - ◇ Advanced Group Representation Theory
  - ◇ Pre-university Bridging Course (twice)
- Marked assignments, exams and quizzes; supervised exams
- Completed Tutor and Demonstrator Development program, 2004

Volunteer Experience

- **Treasurer**
  Postgraduate Mathematics Society (PI), University of Sydney, 2005-6

- **Courses and Careers Day**, University of Sydney, 2005
  Worked on the Mathematics desk, answering students' questions about careers and courses in mathematics.

- **'Science in the City'**, Australian Museum 2004 & 2005
  Represented the University of Sydney for a day at the museum promoting mathematics and science to children.

- **School Tutoring Program Volunteer**, University of Sydney, 2004
  Tutored a student from a disadvantaged local high school in mathematics and English.